Why Do We Add 0.5?

Understanding Continuity Correction

Po Hu

Independent Researcher [1]

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ABSTRACT. Textbooks usually tell us to add 0.5 when a normal curve is used to approximate a discrete distribution. The rule is easy to state, but the reason behind it is rarely explained. This paper takes up two questions: why 0.5, and whether 0.5 is always the most accurate choice. The answer to the first comes from a short sum-to-integral calculation. Since the integers sit one unit apart, shifting the boundary by half a step removes the first lattice error. The answer to the second is different. In one binomial example, a correction of 0.9 gives a much smaller error at a chosen cutoff, and a Poisson example shows the same pattern. The article also notes that there is no fixed sample size that makes an approximation "large." In short, 0.5 is the natural half-step correction, but it is not always the best correction at every point.

KEYWORDS. continuity correction; lattice distribution; normal approximation; skewness; approximation error

1 The question behind the rule

Many probability courses give the following rule. Let X be an integer-valued random variable with mean μ and standard deviation σ. Then

$$P(X \le k) \approx \Phi\left(\frac{k + 0.5 - \mu}{\sigma}\right), \tag{1}$$

where Φ is the standard normal distribution function. When I first met this rule, I simply remembered the number 0.5. But later I asked two questions. Why exactly 0.5? Is it always the most accurate choice?

The two questions sound similar, but they are not the same. The first is about the lattice, which here just means a set of equally spaced points, such as the integers. A discrete variable lives on these points, while a normal curve is continuous. The second is about the shape of the distribution. Even after we put the boundary in the right place, the distribution may still be skewed, and other small errors may remain.

That difference is the main idea of the article. The number 0.5 fixes the first gap between the lattice and the curve. It cannot fix every other source of error. The underlying theory is classical, and standard books and papers are listed in[1, 2, 3, 4].

[1] hoopoe163@gmail.com

2 Why half a step?

2.1. Including k does not choose 0.5

A common classroom explanation starts with the fact that X takes only integer values. We rewrite

$$X \le k \qquad \text{as} \qquad X < k + 0.5.$$

The new boundary includes k but not $k+1$. This is correct and easy to remember. Still, it does not explain why we must add 0.5.

In fact, for every $0 < \theta < 1$,

$$\{X \le k\} = \{X < k + \theta\}. \tag{2}$$

So 0.1, 0.5, and 0.9 all include k but not $k+1$. However, they give different answers when we put them into the normal approximation. Therefore, "include k" only tells us to put the boundary somewhere between k and $k+1$. It does not tell us to choose the middle point.

The first reason for using $k+0.5$ is symmetry. It is exactly halfway between the two nearest integers:

$$k + \frac{1}{2} = \frac{k + (k+1)}{2}.$$

It does not favor either side. So it is the most natural place for the boundary. But "most natural" does not always mean "smallest error." To see why 0.5 is special, we need to look at what happens when a sum is replaced by an integral.

2.2. The first-order error chooses 0.5

Near the center of a lattice distribution, the local central limit theorem gives

$$P(X = j) \approx \frac{1}{\sigma}\phi\left(\frac{j-\mu}{\sigma}\right),$$

where ϕ is the standard normal density. Adding the probabilities up to k is then like using a sum to approximate an integral. The Euler–Maclaurin formula gives a half-weight to the last point. As a result,

$$P(X \le k) \approx \Phi(x) + \frac{1}{2\sigma}\phi(x), \qquad x = \frac{k-\mu}{\sigma}. \tag{3}$$

On the other hand,

$$\Phi\left(x + \frac{c}{\sigma}\right) = \Phi(x) + \frac{c}{\sigma}\phi(x) + O(\sigma^{-2}).$$

Only $c = 1/2$ matches the extra term in (3). So the half-step is not just an easy rule. It removes the first error caused by the lattice. But it does not remove skewness or all the other errors.

3 A case where 0.9 wins

Consider

$$X \sim \mathrm{Bin}(1600, 0.75).$$

Its mean and standard deviation are

$$\mu = 1200, \qquad \sigma = \sqrt{300} \approx 17.320508.$$

Suppose we want to find $P(X \le 1240)$. A direct calculation gives

$$P(X \le 1240) = \sum_{j=0}^{1240} \binom{1600}{j} (0.75)^j (0.25)^{1600-j} \approx 0.990888693836110. \tag{4}$$

Now replace the usual 0.5 by a general value c:

$$A(c) = \Phi\left(\frac{1240 + c - 1200}{\sqrt{300}}\right).$$

Table 1 compares four simple choices.

Table 1: Four normal approximations to $P(X \le 1240)$ for $X \sim \mathrm{Bin}(1600, 0.75)$.

c	$A(c)$	absolute error
0	0.989539332331	0.001349361505
0.1	0.989698310077	0.001190383759
0.5	0.990313340334	0.000575353502
0.9	0.990896041144	**0.000007347309**

The usual correction clearly helps. Still, if we compare absolute errors, 0.9 is about 78 times more accurate than 0.5. This was the calculation that made me question the fixed rule in textbooks. Of course, 0.9 does not become a new rule. The example only shows that half a lattice step cannot remove every error.

3.1. The best correction at one cutoff

For $X \sim \mathrm{Bin}(n, p)$, write

$$F_{n,p}(k) = P(X \le k), \qquad \sigma = \sqrt{np(1-p)}.$$

Now ask a different question. Which value of c makes the normal approximation equal to the exact probability at this cutoff? Solving

$$F_{n,p}(k) = \Phi\left(\frac{k + c' - np}{\sigma}\right)$$

gives

$$\boxed{c'(n, p, k) = np - k + \sigma\Phi^{-1}\big(F_{n,p}(k)\big).} \tag{5}$$

This is the best correction at one point, or the *pointwise optimal correction.* Formula (5) uses the exact probability, so it is not a faster way to calculate the answer. It is only a way to check what is happening at this cutoff.

For the example above,

$$c'(1600, 0.75, 1240) \approx 0.894818617.$$

This explains why 0.9 works so well. There is also a more important fact. The value of c' changes with k, even when n and p stay the same. No fixed number can be best at every cutoff.

We now have two meanings of "best." The value 0.5 puts the boundary halfway between two lattice points and removes the first lattice error. The value $c'(n, p, k)$ is best for one distribution at one cutoff. These two values answer different questions.

4 Why skewness moves the answer

The normal curve is symmetric, but a binomial distribution usually is not. Its skewness is measured by

$$\frac{1-2p}{\sqrt{np(1-p)}}.$$

When $p = 1/2$, the skewness is zero. When $p \neq 1/2$, the two sides of the mean no longer have the same shape. A normal curve can have the same mean and variance, but it cannot show this difference perfectly.

The half-step only fixes the lattice boundary. It does not make a skewed distribution symmetric. So at different cutoffs, the normal curve may need to move a little to the left or right. The best value c' moves too. In our example, $c' \approx 0.894819$. That is why 0.9 is much closer to the exact answer than 0.5.

Again, this does not give us a new 0.9 rule. The change comes from both skewness and the position of the cutoff. More advanced methods can estimate this change, but we do not need their long formulas for the main idea [3, 4].

5 The same idea for Poisson variables

The same idea is not limited to the binomial distribution. Suppose

$$X \sim \mathrm{Pois}(\lambda).$$

This variable also lives on the integer lattice. Its mean and variance are both λ, so the usual normal approximation is

$$P(X \leq k) \approx \Phi\left(\frac{k + 0.5 - \lambda}{\sqrt{\lambda}}\right).$$

Again, 0.5 appears because two nearest values are one unit apart. It does not come from a special version of the central limit theorem.

A Poisson distribution is skewed to the right, so 0.5 still may not give the smallest error at every cutoff. We can define the same kind of best value:

$$c' = \lambda - k + \sqrt{\lambda}\,\Phi^{-1}\big(P(X \le k)\big).$$

For example, if $X \sim \text{Pois}(10)$ and $k = 10$, then

$$P(X \le 10) \approx 0.583039750, \qquad c' \approx 0.663054.$$

We need the exact probability to find c', so this is not a faster method. It only helps us see the same pattern again. Half a step puts the boundary in the middle, while skewness may move the best value somewhere else.

If the distance between lattice points is h, the natural correction is $h/2$. A variable that takes only even integers has distance two, so its half-step is one. For a continuous variable, or a variable that is not on a lattice, adding half a unit has no clear meaning. A central limit theorem may still give a normal limit, but the possible values of the variable decide whether a continuity correction is needed.

6 How large is a large sample?

There is no magic rule such as $n \ge 30$. A central limit theorem tells us what happens when n becomes very large. It does not give one useful cutoff for every problem. Under some common conditions, the Berry–Esseen theorem shows that the error often has size $n^{-1/2}$ [5]. But the constant, the distribution, and the place where we cut the tail still matter.

The words "large sample" become useful only after we choose how much error we can accept. For $X \sim \text{Bin}(n, p)$, let

$$E_n(p) = \max_{0 \le k \le n} \left| P(X \le k) - \Phi\left(\frac{k + 0.5 - np}{\sqrt{np(1-p)}} \right) \right|. \tag{6}$$

This quantity gives the largest error over all cutoffs. For $p = 0.75$, an error below 0.01 needs about $n = 60$. An error below 0.005 needs about $n = 240$. An error close to 0.001 needs about $n = 6000$.

Better methods can make the error smaller, but they also need more conditions and more work. Our aim here is simpler. We only want to know what 0.5 really fixes. The common conditions $np \ge 5$ and $n(1-p) \ge 5$ are useful warning lines, not promises. If we study a tail probability or want a very small error, we may need a much larger sample.

In class, I would still write 0.5. Beside it, I would add the words "half a lattice step." If a student asked, "Is it always best?" I would answer, "No, but it is the right place to start." In our example with $k = 1240$, the best next step happens to be almost 0.9.

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